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BARE-DIE SMART BRIDGE CONNECTED WITH COPPER PILLARS FOR SYSTEM-IN-PACKAGE APPARATUS

Abstract

A system-in-package apparatus includes a semiconductive bridge that uses bare-die pillars to couple with a semiconductive device such as a processor die. The apparatus achieves a thin form factor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. patent application Ser. No. 17/716,958, filed Apr. 8, 2022, which is a continuation of pending U.S. patent application Ser. No. 17/555,219, filed Dec. 17, 2021, which is a continuation of U.S. patent application Ser. No. 16/349,170, filed on May 10, 2019, now U.S. Pat. No. 11,270,941, issued Mar. 8, 2022, which is a U.S. National Phase Application under 35 U.S.C. § 371 of International Application No. PCT/US2016/069176, filed Dec. 29, 2016, entitled “BARE-DIE SMART BRIDGE CONNECTED WITH COPPER PILLARS FOR SYSTEM-IN-PACKAGE APPARATUS,” which are hereby incorporated by reference in their entirety and for all purposes.

FIELD

[0002] This disclosure relates to system-in-package configurations where a bare die semiconductive connector is coupled with copper pillars between two devices.

BACKGROUND

[0003] Package miniaturization poses device-integration challenges, where thin-profile apparatus are useful, but interconnections both active and passive devices require physical protection and heat management while miniaturizing the package.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Disclosed embodiments are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings where like reference numerals may refer to similar elements, in which:

[0005] FIG. 1 is a cross-section elevation of a system-in-package apparatus that includes a semiconductive bridge according to an embodiment;

[0006] FIG. 1A is a cross-section elevation of the system-in-package apparatus depicted in FIG. 1 during assembly according to an embodiment;

[0007] FIG. 1B is a cross-section elevation of the system-in-package apparatus depicted in FIG. 1 after further processing of the structure depicted in FIG. 1A according to an embodiment;

[0008] FIG. 1C is a cross-section elevation of the system-in-package apparatus depicted in FIG. 1 after further process of the structure depicted in FIG. 1B according to an embodiment;

[0009] FIG. 1D is a cross-section elevation of the system-in-package apparatus depicted in FIG. 1 after further process of the structure depicted in FIG. 1C according to an embodiment;

[0010] FIG. 2 is a cross-section elevation of a system-in-package apparatus that includes a redistribution layer and at least a semiconductive bridge and a first integrated circuit die according to an embodiment;

[0011] FIG. 2C is a cross-section elevation of the system-in-package apparatus depicted in FIG. 2

after further process of the structure depicted, for example in FIGS. **1A** and **1B** according to an embodiment;

[0012] FIG. **2D** is a cross-section elevation of the system-in-package apparatus depicted in FIG. **2** after further process of the structure depicted in FIG. **2C** according to an embodiment;

[0013] FIG. **3** is a cross-section elevation of a system-in-package apparatus that includes at least one of a redistribution layer and a semiconductive bridge that includes through-silicon vias according to an embodiment;

[0014] FIG. **4** is a cross-section elevation of a system-in-package apparatus that includes a plurality of semiconductive bridges according to an embodiment;

[0015] FIG. **5** is a process flow diagram that illustrates assembly of a system-in-package that includes at least one semiconductive bridge that is coupled to interconnect pillars according to an embodiment; and

[0016] FIG. **6** is included to show an example of higher level device applications for the disclosed embodiments.

DETAILED DESCRIPTION

[0017] Disclosed embodiments include bare die smart connectors that use a semiconductive bridge that is affixed in a mass such as a molding compound. The smart connector is coupled to interconnect pillars for coupling a semiconductive device such as a processor.

[0018] FIG. **1** is a cross-section elevation of a system-in-package apparatus **100** that includes a semiconductive bridge **10** according to an embodiment. The semiconductive bridge **10** may be referred to as a smart die connector **10**. The semiconductive bridge **10** may be referred to as a bare die silicon bridge **10**.

[0019] The semiconductive bridge **10** is affixed in a mass **110** such as an encapsulation material **110**. The semiconductive bridge **10** includes an active surface **112** and a backside surface **114**. The mass **110** includes a die side **116** and a land side **118**. In an embodiment, the backside surface **114** is fully enclosed in the mass **110**.

[0020] In an embodiment, the mass **110** is a molding compound that is useful for encapsulating semiconductive devices such as the semiconductive bridge **10**. In an embodiment, the mass **110** is a molding compound such as a thermally cured resin material that is useful for encapsulating semiconductive devices such as the semiconductive bridge **10**.

[0021] The system-in-package (SiP) apparatus **100** also includes an interconnect package **13** that is also affixed in the mass **110**. In an embodiment, the interconnect package **13** is a laminated structure **13** that provides interconnect-and trace interconnection (as illustrated in FIG. **2D**) between the die side **116** and the land side **118**. In an embodiment, the interconnect package **13** is a through-package via structure **13** that includes via bars (as illustrated in FIG. **2C**) that pass straight through the interconnect package **13** between the die side **116** and the land side **118**. In an embodiment, the interconnect package **13** is made from organic materials such as FR4 construction. In an embodiment, the interconnect package **13** is made from semiconductive materials. In an embodiment, the interconnect package **13** is made from inorganic materials such as a glass construction.

[0022] In an embodiment, a semiconductive device **11** such as a processor logic die **11** is affixed in a capping material **120** such as a mold cap **120**. In an embodiment, the capping material **120** is an optically cured resin. In an embodiment, the capping material is a thermally cured resin of a different quality from the mass **110**. The semiconductive device **11** may also be referred to as an integrated circuit (IC) die **11**. In an embodiment, the processor logic die **11** is a processor manufactured by Intel Corporation of Santa Clara, California. Electronic communication by the semiconductive device **11** with the semiconductive bridge **10** is facilitated by a first plurality of interconnect pillars, one of which is indicated by reference numeral **121**. The semiconductive device **11** is also coupled to the interconnect package **13** by a third plurality of interconnect pillars, one of which is indicated by the reference numeral **123**.

[0023] In an embodiment, the semiconductive device **11** is a first semiconductive device **11**, and a second semiconductive device **12** such as a memory die **12** is affixed in the capping material **120**. In an embodiment, the second semiconductive device **12** is a memory die manufactured by IM Flash technologies of Lehi, Utah. In a memory-die embodiment, the second semiconductive device **12** may also be referred to as an IC memory die **12**. Electronic communication by the second semiconductor device **12** with the semiconductive bridge **10** is facilitated by a second plurality of interconnect pillars, one of which is indicated by reference numeral **122**.

[0024] In an embodiment, the interconnect package **13** is a first interconnect package **13** and a second interconnect package **14** is also affixed in the mass **110**. In an embodiment, the second interconnect package **14** is a laminated structure **14** that provides interconnect-and-trace interconnection between the die side **116** and the land side **118**. In an embodiment, the second interconnect package **14** is a through-package via structure **14** that includes via bars between the die side **116** and the land side **118**.

[0025] In an embodiment, the SiP apparatus **100** includes a passive component **15** such as a diode **15**. In an embodiment, the passive component **15** is a balun **15** and the second semiconductive device **12** is a baseband processor that is assisted by the balun **15**. Electronic communication by passive component **15** with the semiconductive bridge **10** is facilitated by a fifth plurality of interconnect pillars, one of which is indicated by reference numeral **125**.

[0026] A capping material **120** is provided to cover the devices that are coupled to the semiconductive bridge **10**. In an embodiment, the capping material is a mold cap compound.

[0027] FIG. **1A** is a cross-section elevation **101** of the SiP apparatus **100** depicted in FIG. **1** during assembly according to an embodiment. Cartesian references are given in $-Z$ and X as the structure depicted in FIG. **1A** will be vertically inverted after further processing. A release layer **126** is provided, to which the semiconductive bridge **10** is mounted in a flipped configuration to the release layer **126**. Additionally, the first interconnect package **13** is also positioned on the release layer **126** in an embodiment. Additionally, the second interconnect package **14** is also positioned on the release layer **126** in an embodiment.

[0028] FIG. **1B** is a cross-section elevation **102** of the SiP apparatus **100** depicted in FIG. **1** after further processing of the structure depicted in FIG. **1A** according to an embodiment. A mass **110** has been applied to the semiconductive bridge **10**, the first- and second interconnect packages **13** and **14**, respectively, as well as to the release layer **126**. By this process, the articles **10**, **13** and **14** are affixed and are ready to be inverted for further processing.

[0029] FIG. **1C** is a cross-section elevation **103** of the SiP apparatus **100** depicted in FIG. **1** after further process of the structure depicted in FIG. **1B** according to an embodiment. Cartesian references are given in Z and X as the structure depicted in FIG. **1B** has been vertically inverted. The release layer **126**, depicted in FIG. **1B**, has been removed. It can be seen that bond pads are illustrated, which are useful for describing bonding locations both for interconnect pillars as well as electrical bumps.

[0030] FIG. **1D** is a cross-section elevation **104** of the SiP apparatus **100** depicted in FIG. **1** after further process of the structure depicted in FIG. **1C** according to an embodiment. Placement of the first plurality of interconnect pillars **121** is accomplished by growing the pillars **121** in situ upon the plurality of bond pads that are depicted within the footprint **121'** according to an embodiment. For example, electrolytic deposition of a copper-containing material may be accomplished by growing the interconnect pillars **121** through a mask (not illustrated). In an embodiment, electroless deposition of a primer layer upon a given bond pad is done, such as a precious metal film, e.g., gold, followed by electrolytic deposition of interconnect-grade copper. Placement of the second plurality of interconnect pillars **122** is accomplished by growing the pillars **122** in situ upon the plurality of bond pads that are depicted within the footprint **122'** according to an embodiment. Placement of the third plurality of interconnect pillars **123** is accomplished by growing the pillars **123** in situ upon the plurality of bond pads that are depicted within the footprint **123'** according to

an embodiment. Placement of the fourth plurality of interconnect pillars **124** is accomplished by growing the pillars **124** in situ upon the plurality of bond pads that are depicted within the footprint **124'** according to an embodiment. Placement of the fifth plurality of interconnect pillars **125** is accomplished by growing the pillars **125** in situ upon the plurality of bond pads that are depicted within the footprint **125'** according to an embodiment. It is now understandable that each of the interconnect pillar sets may be grown individually, or a subset of the depicted pillars may be established, depending upon a given useful application needed. In an embodiment, all of the depicted pillars are grown in situ simultaneously.

[0031] Reference is again made to FIG. **1**. After processing depicted in any of the previous figures, an electrical bump array is formed on the interconnect packages, one landside bump of which is enumerated with reference numeral **128**. In an embodiment, a board **130** is assembled to the electrical bump array **128**. In particular, the electrical bump array **128** may be referred to as a landside bump array **128**.

[0032] Useful applications of SiP embodiments that contain the semiconductive bridge **10** include a lowered Z-height due to interconnect pillar length such as in a range between about 10 micrometer (micron) and 50 micron. Useful applications of SiP embodiments that contain the semiconductive bridge **10** include a lowered Z-height due to the semiconductive bridge **10** being located at approximately the same Z-location of the interconnect package **13**, and the material qualities of the mass **110** being sufficiently stiff as to preclude the use of a core material.

[0033] In an embodiment, the semiconductive bridge **10** is referred to as a smart bridge **10** where back-end-of-line (BEOL) metallization connects logic in the smart bridge **10** between the first IC device **11** and the second an IC device **12**. In an embodiment, the smart bridge **10** includes BEOL metallization that connects microcontroller logic in the smart bridge **10** between the first IC device **11** and the second an IC device **12**. In an embodiment, the smart bridge **10** includes BEOL metallization that connects external sensor logic in the smart bridge **10** between the first IC device **11** and the second an IC device **12**. In an embodiment, the smart bridge **10** includes BEOL metallization that connects memory controller logic, with no memory functionality in the smart bridge **10**, but the memory controller logic affects communication between the first IC device **11** and a memory IC device **12**. In an embodiment, the smart bridge **10** includes BEOL metallization that contains switching logic such as for power-conservation functionality or such as temperature-control functionality between the first IC device **11** and the second IC device **12**.

[0034] FIG. **2** is a cross-section elevation of a system-in-package apparatus **200** that includes a redistribution layer **20** and at least a semiconductive bridge **10** and a first IC die **11** according to an embodiment. The redistribution layer (RDL) **20** is useful where, in an example embodiment, increased pin count is desired, particularly in regions between e.g., the semiconductive bridge **10** and a given interconnect package **13**. For example, the RDL **20** expands design freedom as the interconnect pillars are not necessarily tied to a give pad position on the semiconductive bridge **10**, nor to a given pad position of a given interconnect package, or both. In a non-limiting illustrative embodiment, it can be seen that first- and third interconnect pillar footprints **121'** and **123'** respectively have a number of interconnect pillars there (three are illustrated by non-limiting example) between that connect from the first IC die **11** to the RDL **20** and for illustrative purposes, and not by necessity, but the three illustrated interconnect pillars are not directly above either of the semiconductive bridge **10** nor the first interconnect package **13**. Similarly, in a non-limiting illustrative embodiment, it can be seen that second- and fourth interconnect pillar footprints **122'** and **124'** respectively have a number of interconnect pillars there between that connect from the second IC die **12** to the RDL **20**. It should be understood that the RDL **20** does not necessarily provide a direct Z-direction contact between any interconnect pillar and e.g. a device directly underlying the interconnect pillar, although such a direct Z-direction contact is not excluded.

[0035] FIG. **2C** is a cross-section elevation **203** of the SiP apparatus **200** depicted in FIG. **2** after further process of the structure depicted, for example in FIGS. **1A** and **1B** according to an

embodiment. Items 2A and 2B are not used. The release layer **126**, depicted in FIG. **1B**, has been removed. It can be seen that bond pads in interconnect packages **13** and **14** are illustrated substantially flush with the land side **118**, but the RDL **20** precludes explicit illustration of bond pads for the semiconductive bridge **10** and the interconnect packages **13** and **14** where the semiconductive bridge **10** and the packages **13** and **14** are substantially flush with the die side **116** of the mass **110**.

[0036] FIG. **2D** is a cross-section elevation **204** of the SiP apparatus **200** depicted in FIG. **2** after further process of the structure depicted in FIG. **2C** according to an embodiment. Placement of the several pluralities of interconnect pillars **121**, **122**, **123**, **124**, and **125** is accomplished by any technique disclosed herein for the embodiments depicted in FIG. **1D**. It can be seen that more interconnect pillars are depicted than just those categorized within the footprints **121'**, **122'**, **123'**, **124'** and **125'** in order to accommodate a higher pin count in an embodiment. In an embodiment, the pin count may be higher or lower, but placement of the several interconnect pillars may be altered to facilitate the RDL **20**.

[0037] It may now be understood that a via-pillar interconnect package **13** may be used as one- or both of the interconnect packages in any given embodiment. It may now be understood that a via-trace interconnect package **14** may be used as one- or both of the interconnect packages in any given embodiment. It may now be understood that a combination of via-pillar interconnect package **13** and a via-trace interconnect package **14** may be used together in any given embodiment.

[0038] FIG. **3** is a cross-section elevation of a system-in-package apparatus **300** that includes at least one of a redistribution layer **20** and a semiconductive bridge **10** that includes through-silicon vias (TSVs), one of which is illustrated with the numeral **310** according to an embodiment. It can be seen that Z-direction geometries have been altered to allow the backside **114** of the semiconductive bridge **10** to be substantially flush with the land side **118** of the mass **110**. This configuration allows for TSVs **310** to be bumped at the level of the landside bump array **128**.

[0039] In an embodiment, the SiP **300** may be configured without the RDL **20** (restricting the interconnect pillars to the enumerated footprints), and the semiconductive bridge **10** provides TSV communication to the land side **118**. In embodiment without an RDL, no interconnect package (**13** nor **14**) is used such that all communication to the land side **118** is through the TSVs **310**. In embodiment without an RDL, no interconnect package (**13** nor **14**) is used such that all communication to the land side **118** is through the TSVs **310**. In embodiment without an RDL, only one interconnect package (e.g. package **13**) is used such that all communication to the land side **118** is in part through the interconnect package **13** and in part through the TSVs **310**.

[0040] In an embodiment including the RDL **20**, no interconnect package (**13** nor **14**) is used such that all communication to the land side **118** is through the TSVs **310**. In embodiment including the RDL **20**, only one interconnect package (e.g. package **13**) is used such that all communication to the land side **118** is in part through the interconnect package **13** and in part through the TSVs **310**.

[0041] FIG. **4** is a cross-section elevation of a system-in-package apparatus **400** that includes a plurality of semiconductive bridges **10** and **16** according to an embodiment. Similarities are seen in the SiP apparatus **400** to previously disclosed embodiments. In an embodiment, the semiconductive bridge **10** is a first semiconductive bridge **10** and the semiconductive bridge **16** is a subsequent semiconductive bridge **16**. The subsequent semiconductive bridge **16** includes an active surface **132** and a backside surface **134**. Where only two semiconductive bridges are present, the subsequent semiconductive bridge **16** may be referred to as a second semiconductive bridge **16**.

[0042] It can be seen that several interconnect pillars couple devices as well as interconnect packages **13**, **14**, and **18** such that electronic communication may be continuous from the IC die **11** to an external device **17** through the several series of interconnect pillars. In an embodiment, the external device **17** is a camera with a lens **17'**. In an embodiment, the external device **17** includes a touch-sensitive display screen **17'**. In an embodiment, the external device **17** includes a user interface **17'**.

[0043] Whereas the SiP apparatus **400** is illustrated as a bare-die semiconductive bridge-coupled apparatus, it is understood that an RDL may be used between the series of interconnect pillars and the die side **116** of the mass **110** as is illustrated in other disclosed embodiments.

[0044] FIG. **5** is a process flow diagram **500** that illustrates assembly of an SiP that includes at least one semiconductive bridge that is coupled to interconnect pillars according to an embodiment.

[0045] At **510**, the process includes attaching a semiconductive bridge and an interconnect package to a release layer.

[0046] At **520**, the process includes affixing the semiconductive bridge and the interconnect package in a mass.

[0047] At **530**, the process includes removing the release layer.

[0048] At **540**, the process includes assembling first- second- and third pluralities of interconnect pillars to the semiconductive bridge.

[0049] At **550**, the process includes assembling a first semiconductive device to the first- and third pluralities of interconnect pillars.

[0050] At **560**, the process includes applying a capping material to cover the first semiconductive device and in contact with the interconnect pillars.

[0051] At **570**, the process includes assembling the SiP, that includes a smart bridge, to a computing system.

[0052] FIG. **6** is included to show an example of a higher level device application for the disclosed embodiments. In an embodiment, a computing system **600** includes, but is not limited to, a desktop computer. In an embodiment, a system **600** includes, but is not limited to a laptop computer. In an embodiment, a system **600** includes, but is not limited to a netbook. In an embodiment, a system **600** includes, but is not limited to a tablet. In an embodiment, a system **600** includes, but is not limited to a notebook computer. In an embodiment, a system **600** includes, but is not limited to a personal digital assistant (PDA). In an embodiment, a system **600** includes, but is not limited to a server. In an embodiment, a system **600** includes, but is not limited to a workstation. In an embodiment, a system **600** includes, but is not limited to a cellular telephone. In an embodiment, a system **600** includes, but is not limited to a mobile computing device. In an embodiment, a system **600** includes, but is not limited to a smart phone. In an embodiment, a system **600** includes, but is not limited to an internet appliance. Other types of computing device may be configured with the microelectronic device that includes a system-in-package apparatus with a semiconductive bridge embodiment.

[0053] In some embodiments, the system-in-package apparatus with a semiconductive bridge embodiment **600** includes a system on a chip (SOC) system.

[0054] In an embodiment, the processor **610** has one or more processing cores **612** and **612N**, where **612N** represents the Nth processor core inside processor **610** where N is a positive integer. In an embodiment, the electronic device system **600** using a system-in-package apparatus with a semiconductive bridge embodiment that includes multiple processors including **610** and **605**, where the processor **605** has logic similar or identical to the logic of the processor **610**. In an embodiment, the processing core **612** includes, but is not limited to, pre-fetch logic to fetch instructions, decode logic to decode the instructions, execution logic to execute instructions and the like. In an embodiment, the processor **610** has a cache memory **616** to cache at least one of instructions and data for the SiP device system **600**. The cache memory **616** may be organized into a hierarchal structure including one or more levels of cache memory.

[0055] In an embodiment, the processor **610** includes a memory controller **614**, which is operable to perform functions that enable the processor **610** to access and communicate with memory **630** that includes at least one of a volatile memory **632** and a non-volatile memory **634**. In an embodiment, the processor **610** is coupled with memory **630** and chipset **620**. The processor **610** may also be coupled to a wireless antenna **678** to communicate with any device configured to at least one of transmit and receive wireless signals. In an embodiment, the wireless antenna interface

678 operates in accordance with, but is not limited to, the IEEE 802.11 standard and its related family, Home Plug AV (HPAV), Ultra Wide Band (UWB), Bluetooth, WiMax, or any form of wireless communication protocol.

[0056] In an embodiment, the volatile memory **632** includes, but is not limited to, Synchronous Dynamic Random Access Memory (SDRAM), Dynamic Random Access Memory (DRAM), RAMBUS Dynamic Random Access Memory (RDRAM), and/or any other type of random access memory device. The non-volatile memory **634** includes, but is not limited to, flash memory, phase change memory (PCM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), or any other type of non-volatile memory device.

[0057] The memory **630** stores information and instructions to be executed by the processor **610**. In an embodiment, the memory **630** may also store temporary variables or other intermediate information while the processor **610** is executing instructions. In the illustrated embodiment, the chipset **620** connects with processor **610** via Point-to-Point (PtP or P-P) interfaces **617** and **622**. Either of these PtP embodiments may be achieved using a system-in-package apparatus with a semiconductive bridge embodiment as set forth in this disclosure. The chipset **620** enables the processor **610** to connect to other elements in the SiP device system **600**. In an embodiment, interfaces **617** and **622** operate in accordance with a PtP communication protocol such as the Intel® QuickPath Interconnect (QPI) or the like. In other embodiments, a different interconnect may be used.

[0058] In an embodiment, the chipset **620** is operable to communicate with the processor **610**, **605N**, the display device **640**, and other devices **672**, **676**, **674**, **660**, **662**, **664**, **666**, **677**, etc. The chipset **620** may also be coupled to a wireless antenna **678** to communicate with any device configured to at least do one of transmit and receive wireless signals.

[0059] The chipset **620** connects to the display device **640** via the interface **626**. The display **640** may be, for example, a liquid crystal display (LCD), a plasma display, cathode ray tube (CRT) display, or any other form of visual display device. In an embodiment, the processor **610** and the chipset **620** are merged into a single SOC. Additionally, the chipset **620** connects to one or more buses **650** and **655** that interconnect various elements **674**, **660**, **662**, **664**, and **666**. Buses **650** and **655** may be interconnected together via a bus bridge **672**. In an embodiment, the chipset **620** couples with a non-volatile memory **660**, a mass storage device(s) **662**, a keyboard/mouse **664**, and a network interface **666** by way of at least one of the interface **624** and **674**, the smart TV **676**, and the consumer electronics **677**, etc.

[0060] In an embodiment, the mass storage device **662** includes, but is not limited to, a solid state drive, a hard disk drive, a universal serial bus flash memory drive, or any other form of computer data storage medium. In one embodiment, network interface **666** is implemented by any type of well-known network interface standard including, but not limited to, an Ethernet interface, a universal serial bus (USB) interface, a Peripheral Component Interconnect (PCI) Express interface, a wireless interface and/or any other suitable type of interface. In one embodiment, the wireless interface operates in accordance with, but is not limited to, the IEEE 802.11 standard and its related family, Home Plug AV (HPAV), Ultra Wide Band (UWB), Bluetooth, WiMax, or any form of wireless communication protocol.

[0061] While the modules shown in FIG. **6** are depicted as separate blocks within the SiP apparatus in a computing system **600**, the functions performed by some of these blocks may be integrated within a single semiconductor circuit or may be implemented using two or more separate integrated circuits. For example, although cache memory **616** is depicted as a separate block within processor **610**, cache memory **616** (or selected aspects of **616**) can be incorporated into the processor core **612**.

[0062] Where useful, the computing system **600** may have an outer shell that is part of the several land side board embodiments that would be attached at the bump array **128** described in this disclosure. In FIG. **1**, a board **130** is coupled to the electrical bump array **128**. In an embodiment,

an outer shell **131** is an electrically insulated structure on the board **130** that also provides physical protection for the SiP apparatus **100**.

[0063] It may now be understood that a board **130** embodiment may be applied to each illustrated and described electrical bump array **128**.

[0064] To illustrate the memory-die stacked memory module in a system in package apparatus embodiments and methods disclosed herein, a non-limiting list of examples is provided herein:

[0065] Example 1 is a system-in-package apparatus comprising: a semiconductive bridge fixed in a mass, the semiconductive bridge including an active surface and a backside surface, and the mass including a die side and a land side; first- and second pluralities of interconnect pillars extending from the active surface; an interconnect package fixed in the mass, wherein the interconnect package communicates from the die side to the land side; a third plurality of interconnect pillars disposed on the interconnect package at the die side; a first semiconductive die coupled to the first- and third pluralities of interconnect pillars; a second semiconductive die coupled to the second plurality of interconnect pillars; and wherein the first and second semiconductive dice are affixed in a capping material, and wherein the capping material contacts the first- second- and third pluralities of interconnect pillars.

[0066] In Example 2, the subject matter of Example 1 optionally includes wherein the interconnect package is a first interconnect package, further including: a second interconnect package fixed in the mass, wherein the second interconnect package communicates from the die side to the land side; a fourth plurality of interconnect pillars disposed on the second interconnect package at the die side, wherein the second- and fourth pluralities of interconnect pillars are coupled to the second semiconductive die, and wherein the capping material contacts the fourth plurality of interconnect pillars.

[0067] In Example 3, the subject matter of any one or more of Examples 1-2 optionally include a passive device coupled to the semiconductive bridge at a fifth plurality of interconnect pillars that are disposed between the first- and second pluralities of interconnect pillars.

[0068] In Example 4, the subject matter of any one or more of Examples 1-3 optionally include wherein the interconnect package is a first interconnect package, further including: a second interconnect package fixed in the mass, wherein the second interconnect package communicates from the die side to the land side; a fourth plurality of interconnect pillars disposed on the second interconnect package at the die side, wherein the second- and fourth pluralities of interconnect pillars are coupled to the second semiconductive die, and wherein the capping material contacts the fourth plurality of interconnect pillars; wherein the first semiconductive device is a processor device, and wherein the second semiconductive die is a memory device.

[0069] In Example 5, the subject matter of any one or more of Examples 1-4 optionally include an electrical bump array disposed on the land side and coupled to the interconnect package.

[0070] In Example 6, the subject matter of any one or more of Examples 1-5 optionally include a redistribution layer that abuts the several pluralities of interconnect pillars, and wherein the redistribution layer contacts the semiconductive bridge and the interconnect package at a level of the active surface and the die side.

[0071] In Example 7, the subject matter of Example 6 optionally includes wherein the interconnect package is a first interconnect package, further including: a second interconnect package fixed in the mass, wherein the second interconnect package communicates from the die side to the land side; a fourth plurality of interconnect pillars disposed on the second interconnect package at the redistribution layer above the die side, wherein the second- and fourth pluralities of interconnect pillars are coupled to the second semiconductive die through the redistribution layer, and wherein the capping material contacts the fourth plurality of interconnect pillars and the redistribution layer.

[0072] In Example 8, the subject matter of any one or more of Examples 6-7 optionally include a passive device coupled to the redistribution layer by a fifth plurality of interconnect pillars disposed between the first- and second pluralities of interconnect pillars.

[0073] In Example 9, the subject matter of Example 8 optionally includes wherein the passive device is a diode.

[0074] In Example 10, the subject matter of any one or more of Examples 1-9 optionally include wherein the backside surface is fully enclosed in the mass.

[0075] In Example 11, the subject matter of any one or more of Examples 1-10 optionally include wherein the semiconductive bridge includes a through-silicon via, and wherein the backside surface emerges from the mass, further including: an electrical bump array disposed on the land side and coupled to the interconnect package; and wherein the electrical bump array includes an electrical bump on the semiconductive bridge at the backside surface, wherein the through-silicon via is coupled to the electrical bump.

[0076] In Example 12, the subject matter of Example 11 optionally includes wherein the interconnect package is a first interconnect package, further including: a second interconnect package fixed in the mass, wherein the second interconnect package communicates from the die side to the land side; a fourth plurality of interconnect pillars disposed on the second interconnect package at the die side, wherein the second- and fourth pluralities of interconnect pillars are coupled to the second semiconductive die, and wherein the capping material contacts the fourth plurality of interconnect pillars.

[0077] In Example 13, the subject matter of Example 12 optionally includes a passive device coupled to the semiconductive bridge at a fifth plurality of interconnect pillars that are disposed between the first- and second pluralities of interconnect pillars.

[0078] In Example 14, the subject matter of any one or more of Examples 1-13 optionally include an electrical bump array assembled to the interconnect package; and a board assembled to the electrical bump array.

[0079] In Example 15, the subject matter of any one or more of Examples 1-14 optionally include wherein the interconnect package is an interconnect-and-trace connection between the die side and the land side of the mass.

[0080] In Example 16, the subject matter of any one or more of Examples 1-15 optionally include wherein the interconnect package is a through-package via structure connection between the die side and the land side of the mass.

[0081] In Example 17, the subject matter of any one or more of Examples 1-16 optionally include wherein the mass is a thermally cured resin and wherein the capping material is an optically cured resin.

[0082] Example 18 is a system-in-package apparatus comprising: a semiconductive bridge including an active surface and a backside surface; first- and second pluralities of interconnect pillars extending from the active surface; an interconnect package including a die side and a land side, wherein the interconnect package communicates from the die side to the land side; a third plurality of interconnect pillars disposed on the interconnect package at the die side; a first semiconductive die coupled to the first- and third pluralities of interconnect pillars; a second semiconductive die coupled to the second plurality of interconnect pillars; and wherein the first and second semiconductive dice are affixed in a capping material, and wherein the capping material contacts the first- second- and third pluralities of interconnect pillars.

[0083] In Example 19, the subject matter of Example 18 optionally includes wherein the interconnect package is a first interconnect package, further including: a second interconnect package including a die side and a land side that are substantially coplanar with the first interconnect package die side and land side; a fourth plurality of interconnect pillars disposed on the second interconnect package at the die side, wherein the second- and fourth pluralities of interconnect pillars are coupled to the second semiconductive die, and wherein the capping material contacts the fourth plurality of interconnect pillars.

[0084] In Example 20, the subject matter of any one or more of Examples 18-19 optionally include a passive device coupled to the semiconductive bridge at a fifth plurality of interconnect pillars that

are disposed between the first- and second pluralities of interconnect pillars.

[0085] Example 21 is a method of assembling a bridge-containing a system-in-package (SiP) apparatus, comprising: attaching a semiconductive bridge and an interconnect package to a release layer, wherein the semiconductive bridge includes an active surface and a backside surface; affixing the semiconductive bridge and the interconnect package in a mass; removing the release layer; assembling first- and second pluralities of interconnect pillars to the semiconductive bridge; assembling a third plurality of interconnect pillars to the interconnect package; coupling a logic die to first- and third pluralities of interconnect pillars; and affixing the logic die and the pluralities of interconnect pillars in a capping material, wherein the capping material contacts the semiconductive bridge active surface.

[0086] In Example 22, the subject matter of Example 21 optionally includes wherein the interconnect package is a first interconnect package, further including: attaching a second interconnect package to the release layer; affixing the second interconnect package in the mass; assembling a fourth plurality of interconnect pillars to the second interconnect package; coupling a memory die to second- and fourth pluralities of interconnect pillars; and affixing the memory die and the pluralities of interconnect pillars in the capping material.

[0087] In Example 23, the subject matter of Example 22 optionally includes wherein the semiconductive bridge is a first semiconductive bridge, the method further including: attaching a third interconnect package and a second semiconductive bridge to the release layer; affixing the third interconnect package and the second semiconductive bridge in the mass; coupling a user interface to the third interconnect package and the second semiconductive bridge; and affixing the user interface in the capping material.

[0088] In Example 24, the subject matter of Example 23 optionally includes assembling an electrical bump array to the first-second and third interconnect packages at the land side; and assembling a board to the electrical bump array.

[0089] In Example 25, the subject matter of any one or more of Examples 21-24 optionally include assembling an electrical bump array to the interconnect package at the land side; and assembling a board to the electrical bump array.

[0090] Example 26 is a computing system containing a system-in-package (SiP) apparatus, comprising: a semiconductive bridge fixed in a mass, the semiconductive bridge including an active surface and a backside surface, and the mass including a die side and a land side; first- and second pluralities of interconnect pillars extending from the active surface; a first interconnect package fixed in the mass, wherein the first interconnect package communicates from the die side to the land side; a third plurality of interconnect pillars disposed on the interconnect package at the die side; a second interconnect package fixed in the mass, wherein the second interconnect package communicates from the die side to the land side; a fourth plurality of interconnect pillars disposed on the second interconnect package at the die side; a first semiconductive die coupled to the first- and third pluralities of interconnect pillars; a second semiconductive die coupled to the second- and fourth plurality of interconnect pillars; wherein the first and second semiconductive dice are affixed in a capping material, and wherein the capping material contacts the first- second- and third pluralities of interconnect pillars; an electrical bump array coupled to the first- and second interconnect packages at the land side; and a board coupled to the electrical bump array, wherein the board includes an outer shell that provides electrical insulation to the SiP apparatus.

[0091] In Example 27, the subject matter of Example 26 optionally includes a second semiconductive bridge fixed in the mass; a third interconnect package fixed in the mass, wherein the third interconnect package is exposed as both the die side and the land side; and a user interface coupled to the second semiconductive bridge and the third interconnect package.

[0092] The above detailed description includes references to the accompanying drawings, which form a part of the detailed description. The drawings show, by way of illustration, specific embodiments in which the invention can be practiced. These embodiments are also referred to

herein as “examples.” Such examples can include elements in addition to those shown or described. However, the present inventors also contemplate examples in which only those elements shown or described are provided. Moreover, the present inventors also contemplate examples using any combination or permutation of those elements shown or described (or one or more aspects thereof), either with respect to a particular example (or one or more aspects thereof), or with respect to other examples (or one or more aspects thereof) shown or described herein.

[0093] In the event of inconsistent usages between this document and any documents so incorporated by reference, the usage in this document controls.

[0094] In this document, the terms “a” or “an” are used, as is common in patent documents, to include one or more than one, independent of any other instances or usages of “at least one” or “one or more.” In this document, the term “or” is used to refer to a nonexclusive or, such that “A or B” includes “A but not B,” “B but not A,” and “A and B,” unless otherwise indicated. In this document, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Also, in the following claims, the terms “including” and “comprising” are open-ended, that is, a system, device, article, composition, formulation, or process that includes elements in addition to those listed after such a term in a claim are still deemed to fall within the scope of that claim. Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects.

[0095] Method examples described herein can be machine or computer-implemented at least in part. Some examples can include a computer-readable medium or machine-readable medium encoded with instructions operable to configure an electrical device to perform methods as described in the above examples. An implementation of such methods can include code, such as microcode, assembly language code, a higher-level language code, or the like. Such code can include computer readable instructions for performing various methods. The code may form portions of computer program products. Further, in an example, the code can be tangibly stored on one or more volatile, non-transitory, or non-volatile tangible computer-readable media, such as during execution or at other times. Examples of these tangible computer-readable media can include, but are not limited to, hard disks, removable magnetic disks, removable optical disks (e.g., compact disks and digital video disks), magnetic cassettes, memory cards or sticks, random access memories (RAMs), read only memories (ROMs), and the like.

[0096] The above description is intended to be illustrative, and not restrictive. For example, the above-described examples (or one or more aspects thereof) may be used in combination with each other. Other embodiments can be used, such as by one of ordinary skill in the art upon reviewing the above description. The Abstract is provided to comply with 37 C.F.R. § 1.72(b), to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. Also, in the above Detailed Description, various features may be grouped together to streamline the disclosure. This should not be interpreted as intending that an unclaimed disclosed feature is essential to any claim. Rather, inventive subject matter may lie in less than all features of a particular disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description as examples or embodiments, with each claim standing on its own as a separate embodiment, and it is contemplated that such embodiments can be combined with each other in various combinations or permutations. The scope of the invention should be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

Claims

- 1.** A system-in-package apparatus, comprising: a semiconductive bridge in a molding compound, the semiconductive bridge having a top surface, a bottom surface, a first side between the top surface and the bottom surface, and a second side between the top surface and the bottom surface; a first interconnect and a second interconnect laterally adjacent to the first side of the semiconductive bridge; a third interconnect and a fourth interconnect laterally adjacent to the second side of the semiconductive bridge; a redistribution layer over the semiconductive bridge, over the molding compound, and over the first interconnect, the second interconnect, the third interconnect and the fourth interconnect; a first IC device attached to the redistribution layer, the first IC device electrically coupled to the semiconductive bridge, to the first interconnect and to the second interconnect; a second IC device attached to the redistribution layer, the second IC device electrically coupled to semiconductive bridge, to the third interconnect and to the fourth interconnect; a capping material between and in contact with the first IC device and the second IC device; a first plurality of bumps beneath the bottom surface of the semiconductive bridge, the first plurality of bumps within a footprint of the first side and the second side of the semiconductive bridge; a second plurality of bumps beneath the first plurality of interconnects; and a third plurality of bumps beneath the second plurality of interconnects.
- 2.** The system-in-package apparatus of claim 1, further comprising: a second bridge in the molding compound, the second bridge laterally adjacent to one of the first plurality of interconnects or the second plurality of interconnects, wherein the redistribution layer is over the second bridge.
- 3.** The system-in-package apparatus of claim 1, further comprising: a third IC device attached to the redistribution layer, wherein the second bridge couples the second IC device to the third IC device.
- 4.** The system-in-package apparatus of claim 1, further comprising: a third IC over the semiconductive bridge.
- 5.** The system-in-package apparatus of claim 4, wherein the third IC device is between the first IC device and the second IC device.
- 6.** The system-in-package apparatus of claim 4, wherein the third IC is a passive device.
- 7.** The system-in-package apparatus of claim 1, wherein the first interconnect and the second interconnect are spaced at a pitch greater than a pitch of contacts coupling the first IC device to the first interconnect and the second interconnect.
- 8.** The system-in-package apparatus of claim 1, wherein the first interconnect and the second interconnect are in a first laminate package, and the third interconnect and the fourth interconnect are in a second laminate package.
- 9.** The system-in-package apparatus of claim 1, wherein the semiconductor bridge comprises a plurality of through vias therein.
- 10.** The system-in-package apparatus of claim 1, wherein the molding compound is on the bottom surface of the semiconductive bridge.
- 11.** A method of fabricating a system-in-package apparatus, the method comprising: providing a semiconductive bridge in a molding compound, the semiconductive bridge having a top surface, a bottom surface, a first side between the top surface and the bottom surface, and a second side between the top surface and the bottom surface; forming a first interconnect and a second interconnect laterally adjacent to the first side of the semiconductive bridge; forming a third interconnect and a fourth interconnect laterally adjacent to the second side of the semiconductive bridge; forming a redistribution layer over the semiconductive bridge, over the molding compound, and over the first interconnect, the second interconnect, the third interconnect and the fourth interconnect; attaching a first IC device to the redistribution layer, the first IC device electrically coupled to the semiconductive bridge, to the first interconnect and to the second interconnect; attaching a second IC device to the redistribution layer, the second IC device electrically coupled to semiconductive bridge, to the third interconnect and to the fourth interconnect; forming a capping

material between and in contact with the first IC device and the second IC device; forming a first plurality of bumps beneath the bottom surface of the semiconductive bridge, the first plurality of bumps within a footprint of the first side and the second side of the semiconductive bridge; forming a second plurality of bumps beneath the first plurality of interconnects; and forming a third plurality of bumps beneath the second plurality of interconnects.

12. The method of claim 11, further comprising: providing a second bridge in the molding compound, the second bridge laterally adjacent to one of the first plurality of interconnects or the second plurality of interconnects, wherein the redistribution layer is over the second bridge.

13. The method of claim 11, further comprising: attaching a third IC device to the redistribution layer, wherein the second bridge couples the second IC device to the third IC device.

14. The method of claim 11, further comprising: providing a third IC over the semiconductive bridge.

15. The method of claim 14, wherein the third IC device is between the first IC device and the second IC device.

16. The method of claim 14, wherein the third IC is a passive device.

17. The method of claim 11, wherein the first interconnect and the second interconnect are spaced at a pitch greater than a pitch of contacts coupling the first IC device to the first interconnect and the second interconnect.

18. The method of claim 11, wherein the first interconnect and the second interconnect are in a first laminate package, and the third interconnect and the fourth interconnect are in a second laminate package.

19. The method of claim 11, wherein the semiconductor bridge comprises a plurality of through vias therein.

20. The method of claim 11, wherein the molding compound is on the bottom surface of the semiconductive bridge.
